

RONCHI DEI LEGIONARI, Italy

WMO: 161080

Lat: 45.828N

Lon: 13.472E

Elev: 37

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	26.3	7.2	8.0	29.1	11.7	10.0	32.6	23.6	39.4	20.7	39.7	3.2	50	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	91.7	73.2	89.4	72.1	86.4	70.7	76.0	86.6	74.3	84.9	72.8	83.2	6.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	123.3	81.4	71.2	115.3	79.8	69.5	108.4	78.3	39.4	86.5	37.8	84.4	36.4	83.3	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	16.3	13.2	17.8	96.0	5.0	3.0	14.2	98.1	11.3	99.9	8.5	101.5	4.9	103.7
			15.1	79.0	4.7	1.8	11.8	80.3	9.0	81.3	6.4	82.3	3.0	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.0	40.1	41.6	48.1	55.3	63.4	70.9	74.6	74.0	65.9	57.8	49.4	41.6
	DBStd	13.46	5.42	5.56	5.39	5.37	5.02	5.32	4.65	5.16	5.40	5.88	6.69	5.79
	HDD50	1025	311	238	96	15	0	0	0	0	0	10	89	266
	HDD65	3828	773	655	523	294	90	11	3	54	231	467	726	
	CDD50	3571	3	3	38	173	417	627	761	744	476	251	73	5
	CDD65	898	0	0	0	2	41	188	298	282	80	7	0	0
	CDH74	8504	0	0	1	36	379	1772	2963	2745	577	31	0	0
	CDH80	3066	0	0	0	3	69	600	1161	1110	121	2	0	0
(14) Wind	WSAvg	4.4	3.9	4.6	5.0	4.7	4.5	4.4	4.7	4.5	4.4	4.2	3.9	3.9
(15) Precipitation	PrecAvg	46.7	3.0	2.8	3.1	4.0	3.7	3.7	3.6	3.5	4.8	5.8	5.4	3.7
	PrecMax	64.5	10.5	9.5	10.2	9.3	9.6	9.3	6.3	8.5	11.9	12.7	17.1	9.1
	PrecMin	33.4	0.3	0.1	0.0	0.4	0.5	0.9	1.3	0.5	1.5	0.6	0.9	0.2
	PrecStd	8.1	2.3	2.6	2.6	2.2	2.0	1.7	1.5	2.1	2.7	3.4	3.6	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	61.2	69.8	78.6	85.8	93.2	94.9	96.4	87.5	77.4	67.7	58.8
		MCWB	51.8	49.1	53.9	61.7	68.7	74.0	76.4	74.1	70.2	63.0	57.9	53.9
	2%	DB	53.7	57.1	65.4	73.4	81.9	89.4	91.4	92.6	83.2	73.4	64.1	55.4
		MCWB	48.6	47.6	52.6	59.1	66.6	71.9	73.7	72.7	68.4	63.5	57.9	50.2
	5%	DB	51.5	53.9	61.8	70.0	78.5	86.3	89.1	89.5	80.4	70.2	61.2	53.1
		MCWB	46.6	46.4	51.1	56.9	64.3	70.4	72.4	72.0	67.2	62.3	56.1	47.8
	10%	DB	48.6	51.6	58.8	66.5	75.1	83.8	86.2	86.2	77.1	68.0	59.1	50.3
		MCWB	44.6	45.3	49.7	55.5	62.8	69.5	71.0	70.6	65.7	61.0	54.4	46.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	52.1	56.5	63.3	71.3	76.8	78.6	77.9	74.0	68.9	61.9	56.4
		MCDB	55.9	57.1	64.4	74.5	80.6	86.6	89.9	87.1	81.8	71.3	64.7	58.0
	2%	WB	50.6	50.2	54.3	60.6	68.2	73.9	76.2	75.8	71.2	66.5	59.4	52.3
		MCDB	52.4	53.8	61.6	70.2	78.9	85.0	87.3	86.3	78.5	70.6	62.4	54.4
	5%	WB	47.9	48.2	52.8	58.5	65.9	72.2	74.4	74.2	69.2	64.5	57.5	49.2
		MCDB	49.6	52.1	59.1	66.9	75.3	83.4	85.1	84.7	76.6	68.6	60.9	51.0
	10%	WB	45.5	46.2	51.4	56.7	64.0	70.4	72.7	72.6	67.4	62.4	55.5	47.2
		MCDB	47.6	50.0	56.6	64.3	72.3	80.8	83.2	83.0	74.7	66.3	58.5	49.4
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	16.8	18.4	18.2	18.5	19.7	20.5	20.8	19.0	17.5	14.6	14.7
		MCDBR	17.1	21.9	24.9	24.8	24.2	23.6	23.8	24.8	22.7	20.3	15.8	18.0
	5% WB	MCWBR	12.0	14.2	14.9	13.3	11.9	10.9	10.8	10.6	11.4	12.0	10.9	13.1
		MCWBR	10.6	16.0	19.2	21.0	20.8	21.6	22.1	22.3	18.8	14.7	11.8	11.5
(40) Clear-Sky Solar Irradiance	taub		0.342	0.368	0.416	0.460	0.466	0.480	0.468	0.459	0.440	0.419	0.375	0.338
		taud	2.400	2.317	2.192	2.098	2.130	2.142	2.185	2.225	2.240	2.263	2.334	2.431
	Ebn at Noon		238	252	254	253	256	253	254	251	244	229	220	226
		Edh at Noon	23	30	40	48	48	47	45	42	38	32	25	21
(44) All-Sky Solar Radiation	RadAvg		431	706	1101	1447	1765	1995	2026	1742	1269	797	456	376
	RadStd		65	117	142	167	182	119	114	145	114	89	69	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)		+0.81	N/A	N/A	+1.30	+0.79	N/A	-144	-227	+158

Nomenclature: See separate page